

# Optimization, SVM, Convex Clustering Scribe

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## 1 Convex Geometry

### What is a Convex Set?

**Definition 1.1.**  $C$  is called a **convex set** if  $\forall x, y \in C$ , and  $\forall \alpha \in [0, 1]$ , then:

$$\alpha x + (1 - \alpha)y \in C$$

**Definition 1.2.**  $\sum \alpha_i x_i$  with  $\sum \alpha_i = 1$  and  $\alpha_i \geq 0$  is called a **convex combination** of  $x_1, \dots, x_n$ .

**Proposition 1.1.**  $S$  is convex  $\iff$  every convex combination of finitely many points in  $S$  is in  $C$ .

*Proof.*  $\Rightarrow$ : Easy; Induction on  $n$ .

$\Leftarrow$ : From Defn.

□

**Definition 1.3.** Take a set of points  $S = \{x_1, x_2, \dots, x_n\}$ . What is the smallest convex set that contains  $S$ ? This set is named as  $\text{conv}(S)$  or the **convex hull** of  $S$ .

**Proposition 1.2.** Let  $S = \{x_1, \dots, x_n\}$ . Then:

$$\text{conv}(S) = \left\{ \sum_{i=1}^n \theta_i x_i \mid \theta_i \geq 0 \text{ and } \sum_{i=1}^n \theta_i = 1 \right\}$$

*Proof.*  $C = \{\theta_1 x_1 + \dots + \theta_n x_n \mid \theta_i \geq 0, \sum \theta_i = 1\}$  is convex. If  $C'$  is a convex set containing  $S$ , then  $C \subseteq C'$ .  $\tilde{C}$  is smallest. Then  $C \subseteq \tilde{C}$ .  $\therefore C = \tilde{C}$ .  $\square$

**Definition 1.4. Convex Cone:**  $x_1, x_2, \dots, x_n$

$$\text{Cone}(x_1, \dots, x_n) = \left\{ \sum_{i=1}^n \theta_i x_i \mid \theta_i \geq 0 \right\}$$

**Definition 1.5. Primitive Cone:** If  $x_1, \dots, x_k$  are lin. indep. the cone generated by them is called Primitive.

## Some important Lemmas

**Lemma 1.1.**  $C$  is a convex cone in  $\mathbb{R}^m$  generated by fin. many vectors  $a_1, \dots, a_n$ . Then  $C$  can be expressed as the union of fin. many primitive cones.

*Proof.*  $x \in C$ . Take  $I$  to be the minimal subset of  $[n]$  so that  $x \in A_I$  where  $A_I = \{a_{i_1}, \dots, a_{i_k}\}$  and  $x = \sum_{j=1}^k \theta_j a_{i_j}$  and  $\theta_j \geq 0$ . Now from minimality of  $I$ ,  $\theta_j > 0, \forall j \in [k]$ .

We claim  $A_I$  is lin. indep. If not  $\exists c_1, \dots, c_k$  not all 0:

$$0 = \sum_{j=1}^k c_j a_{i_j}$$

$$\sum_{j=1}^k (\theta_j - t c_j) a_{i_j} = x \quad \forall t \in \mathbb{R}$$

Let  $S_1 = \{j \in [k] \mid c_j \neq 0\}$ .  $j^*$  be so that  $\frac{\theta_{j^*}}{c_{j^*}}$  is min. with  $j \in S_1$ . Thus,  $x$  can be represented as a non-negative combination of less than  $|I|$  many  $a_i$ 's contradiction.  $\square$

**Lemma 1.2.** Every primitive cone is closed.

*Proof.* Let a cone be gen. by  $\{a_1, \dots, a_k\}$  where  $a_1, \dots, a_k$  are lin. indep.,  $a_i \in \mathbb{R}^m$ . Consider  $P_0 \subseteq \mathbb{R}^k$  and std. basis  $\{e_1, \dots, e_k\}$ .  $C_0 = \text{cone}\{e_1, \dots, e_k\}$  is the non-negative orthant which is closed. Define  $f : \mathbb{R}^k \rightarrow \mathbb{R}^m$

$$f(x_1, \dots, x_k) = x_1 a_1 + x_2 a_2 + \dots + x_k a_k$$

Now as  $\{a_i\}_{i=1}^k$  is lin. indep.,  $f$  is inj. (cont.).  $f(C_0) = C$ . Since  $f$  is an injective linear mapping,  $f : S \rightarrow f(S)$  is a bijection with a continuous inverse. So,  $C$  is closed.  $\square$

**Proposition 1.1.** Every finitely gen. cone is closed.

*Proof.* Trivial from previous lemmas.  $\square$

## 2 Farkas Lemma

**Theorem 2.1.**  $A \in \mathbb{R}^{m \times n}, \underline{b} \in \mathbb{R}^m$ . Then exactly one of the following happens:

1.  $\exists \underline{x} \in \mathbb{R}^n$  so that  $A\underline{x} = \underline{b}$  and  $\underline{x} \geq 0$
2.  $\exists \underline{y} \in \mathbb{R}^m$  so that  $\underline{y}^T A \leq 0^T$  and  $\underline{y}^T \underline{b} > 0$

**Remark:** Geometric Interpretation of Farkas Lemma:  $A = [a_1 : \dots : a_n]$ . Then essentially Farkas Lemma means that either the vector  $\underline{b}$  will be inside the cone generated by the column vectors of  $A$ , or there will exist a separating hyper-plane through origin separating  $\underline{b}$  and the column vectors into separate half-spaces.

*Proof.* Claim: Both can't hold together.

$$\begin{aligned} A\underline{x} &= \underline{b} \\ \underline{y}^T A\underline{x} &= \underline{y}^T \underline{b} > 0 \end{aligned}$$

However,  $(\underline{y}^T A)\underline{x} \leq 0^T \underline{x} \leq 0$ . Contradiction.

Let  $C = \text{cone}\{a_1, a_2, \dots, a_n\}$  where  $a_i$  denotes the  $i$ th column vector of matrix  $A$ . Let  $\underline{b} \notin C$

Claim: There is a nearest point of  $\underline{b}$  in  $C$  (It might not be unique)

If not, then take any  $r > 0 \exists$  a point  $\underline{x}_r \in C$  such that  $\|\underline{x}_r - \underline{b}\| < r$ . Thus take a sequence converging to 0 and consider the corresponding  $\underline{x}_r$ , they converge to  $\underline{b}$ . But note that  $C$  is closed and  $\underline{b} \notin C$  so we get the contradiction.

Note that  $\langle z, z - b \rangle = 0$  ( Why ? ) Name,  $x - b = y$ , and consider the hyperplane  $y^\perp$ . Note that as  $y'z = 0$ , so  $y'b = y'y < 0$ . Now we claim that  $y'x \geq 0 \forall x \in C$ .

Others=wisem  $\exists x \in C$  such that  $y'x < 0$  Take  $\alpha$  sufficiently small in  $(0, 1)$ . note that

$$\|b - (\alpha x + (1 - \alpha)z)\|^2 = \|y\|^2 + \alpha(\alpha\|z - x\|^2 + 2y'x)$$

. Note that making  $\alpha$  sufficiently small we get contradiction to the fact that  $z$  is the nearest point to  $b$  in  $C$ .  $\square$

**Theorem 2.2 (Modified Farkas Lemma).**  $A^{m \times n}, \underline{b} \in \mathbb{R}^m$ .

$Ax = b$  has nng solutions iff  $\forall y \in \mathbb{R}^n$  such that  $y'A \geq 0, y'b \geq 0$

*Proof.* Immediate from the main statement of Farkas Lemma.  $\square$

**Theorem 2.3 (Farkas Lemma Inequality Form).**  $A^{m \times n}, \underline{b} \in \mathbb{R}^m$

$Ax \leq b$  has non negative solution iff whenever  $y \geq 0, y'A \geq 0'$ , we must have  $y'b \geq 0$

*Proof.* Consider  $\bar{A} = [A | I_m]$  and apply Modified Farkas Lemma.  $\square$

### 3 Linear Programming

Consider, the optimization problem:

$$\begin{aligned} \max c^T x \quad (\text{or } \min) \\ \text{subj. to } Ax = b, \quad x \geq 0 \end{aligned}$$

This is the **Equational Form**.

$$\begin{aligned} \max c^T x \\ \text{subj. to } Ax \leq b, \quad x \geq 0 \end{aligned}$$

This is the **Canonical Form**.

Introduce slack variables to go from C.F. to E.F.  $Ax = b$  are called "constraints". Any  $x$  satisfying the constraints is called a **feas. soln.**

**Definition 3.1. Basic Feasible Soln:**  $A^{m \times n}$ ,  $m \leq n$ . A feasible sol.  $x$  is called **bfs** if  $\exists K \subseteq [n]$  with  $|K| = m$ , so that:

1.  $A|_K$  is inv.
2.  $x_j = 0$  if  $j \notin K$ .

Note that #BFS is finite ( $\binom{n}{m}$  at most).

**Lemma 3.1.** Consider the LPP,  $\max c^T x$  subject to  $Ax = b, x \geq 0$ . If  $x_0$  is a feas. soln and the obj. fn. is bounded  $\implies \exists \tilde{x}$  bfs so that  $c^T x_0 \leq c^T \tilde{x}$ .

*Proof.* Let  $A = \{x' \mid c^T x_0 \leq c^T x', Ax' = b\}$ . Consider  $\tilde{x} \in A$  so that no. of zeros is highest. If  $\tilde{x}$  is bfs then done. If not, let  $J = \{j \in [n] \mid \tilde{x}_j > 0\}$ .  $A|_J$  is lin. dep.  $\exists v \in \mathbb{R}^{|J|} \setminus \{0\}$  so that  $A|_J v = \underline{0}$ . Make other components 0 to get  $\underline{w}$ :

$$\begin{aligned} A|_J v &= A\underline{w} = \underline{0} \\ A\tilde{x} &= b, \quad A\underline{w} = 0 \\ A(\tilde{x} + t\underline{w}) &= b \end{aligned}$$

Now,  $\underline{w}$  and  $\tilde{x}$  have same nonzero components. Consider  $j$  so that  $w_j \neq 0$  and  $\frac{\tilde{x}_j}{w_j}$  is min. Take  $t^* = -\frac{\tilde{x}_{j_0}}{w_{j_0}}$ .  $x^* = \tilde{x} + t^* \underline{w}$  has more 0 comp. than  $\tilde{x}$ .

$$c^T x^* = c^T \tilde{x} + t^* c^T \underline{w}$$

If  $c^T \underline{w} > 0$ , then note that:

$$c^T(\tilde{x} + t\underline{w}) = c^T \tilde{x} + t c^T \underline{w}$$

Take  $t$  arbitrarily large  $\implies$  Obj. fn unbdd.

$\therefore c^T \underline{w} \leq 0$ . So,  $t < 0 \implies c^T x^* \geq c^T \tilde{x} \geq c^T x_0$ .

This contradicts that  $\tilde{x}$  has the highest no. of 0's. Hence,  $\tilde{x}$  is a **bfs**. □

## Fundamental Thm of LPP

Consider LPP:

$$\begin{aligned} &\max c^T x \\ &\text{Subj. to } Ax = b \\ &x \geq 0 \end{aligned}$$

1. If  $\exists$  at least one feas. soln & obj. fn is bdd. from above in feas. reg. then  $\exists$  one bfs.
2. If  $\exists$  at least one optimal soln, then  $\exists$  a bfs which is optimal.

## What do bfs look like?

**Definition 3.1. Convex Polyhedron:** Intersection of fin. many closed half spaces.

**Definition 3.2. Vertex:**  $P \subseteq \mathbb{R}^n$  polyhedron.  $v \in P$  is called a vertex if  $\exists \underline{a} \neq 0$  so that  $\forall x \in P$ ,

$$\underline{a}^T x < \underline{a}^T v$$

Note that the set of feasible soln is a convex poly.

**Proposition 3.1.**  $v \in P$  is a vertex  $\iff v$  is a bfs of the lin. program.

*Proof.*  $\Rightarrow$ :  $v$  is a vertex of  $P$  indexed by vector  $\underline{a}$  say. Consider the LPP:

$$\begin{aligned} \max \underline{a}^T x \\ Ax = b, \quad x \geq 0 \end{aligned}$$

Then  $v$  is the optimal sol. By Fund. Thm  $v$  is a bfs of  $Ax = b$ . □

$$c^T x^* = c^T \tilde{x} + t^* c^T w$$

If  $c^T w > 0$ , then note that:

$$c^T(\tilde{x} + tw) = c^T \tilde{x} + t c^T w$$

Take  $t$  arbitrarily large  $\Rightarrow$  obj. fn unbdd.

$\therefore c^T w \leq 0$ . So, with  $t < 0$ :

$$c^T x^* \geq c^T \tilde{x} \geq c^T x_0$$

This contradicts that  $\tilde{x}$  has the highest no. of 0's. Hence,  $\tilde{x}$  is a **bfs**.

$\Leftarrow$ : Define,

$$d_j = \begin{cases} 0 & \text{if } v_j \neq 0 \\ -1 & \text{if } v_j = 0 \end{cases}$$

$x \in P$ . Then if  $x$  and  $v$  don't have same positions of zero comp.,  $d^T x < 0 = d^T v$ .

Let  $x$  and  $v$  have same positions of zeroes. Then  $d^T x = d^T v$ . But again, then as  $v$  is bfs, the nonzero comp. are uniquely determined ( $A_v^{-1}b$ ). Thus,  $x = v$ . Thus,  $v$  is a vertex of  $P$  indexed by  $d$ .

## 4 Simplex Method

## 5 Duality

| Primal               | Dual                    |
|----------------------|-------------------------|
| $\max c^T x$         | $\min b^T y$            |
| subj. to $Ax \leq b$ | subj. to $A^T y \geq c$ |
| $x \geq 0$           | $y \geq 0$              |

**Why does this make sense?**

....

**Theorem 5.1 (Weak Duality Theorem).**  $x, y$  feas. soln. to the primal and dual LPP respectively.

$$c^T x \leq b^T y$$

*Proof.*  $b^T y \geq x^T A^T y \geq x^T c \geq c^T x$  □

But we can do better in case of Linear Programming and later we will see in many cases in Convex Optimization and Optimization in general.

### (Strong) Duality Theorem and proof

...

*Proof.* Let  $c^T x^* = \gamma$ . Note that for the system  $Ax \leq b, c^T x \geq \gamma$ , there exist non-negative solutions. However, for any  $\epsilon > 0$ , the system  $Ax \leq b, c^T x \geq \gamma + \epsilon$  has no non-negative solution.

Define the augmented matrix  $\bar{A}$  and vector  $\bar{b}_\epsilon$  as:

$$\bar{A} = \begin{bmatrix} A \\ -c^T \end{bmatrix}, \quad \bar{b}_\epsilon = \begin{bmatrix} b \\ -\gamma - \epsilon \end{bmatrix}$$

By **Farkas' Lemma**, there exists a vector  $y_\epsilon \geq 0$  such that  $y_\epsilon^T \bar{A} \geq 0^T$  but  $y_\epsilon^T \bar{b}_\epsilon < 0$ . We partition  $y_\epsilon$  as:

$$y_\epsilon = \begin{bmatrix} u_\epsilon \\ z_\epsilon \end{bmatrix}$$

where  $u_\epsilon \geq 0$  and  $z_\epsilon \geq 0$ . Substituting these into the inequalities, we obtain:

1.  $u_\epsilon^T A - z_\epsilon c^T \geq 0^T \implies A^T u_\epsilon \geq z_\epsilon c$
2.  $u_\epsilon^T b - z_\epsilon(\gamma + \epsilon) < 0 \implies u_\epsilon^T b < z_\epsilon(\gamma + \epsilon)$

If  $z_\epsilon = 0$ , then  $A^T u_\epsilon \geq 0$  and  $u_\epsilon^T b < 0$ , contradiction as both the vectors are nng. Thus,  $z_\epsilon > 0$ . Let  $y = \frac{u_\epsilon}{z_\epsilon}$ . Then:

$$A^T y \geq c \quad \text{and} \quad y^T b < \gamma + \epsilon$$

Since  $y$  is a feasible solution for the dual problem (D), the minimum value of the dual is bounded by  $\gamma + \epsilon$  for all  $\epsilon > 0$ . Consequently, the minimum value of the dual LPP is exactly  $\gamma$ .  $\square$

## 6 Convex Optimization

**Definition 6.1 (Convex Optimization Problem).** Minimize  $f_0(x)$   
subject to  $f_i(x) \leq 0 \quad i = 1, \dots, m$ , and  $Ax = b$ , where  $f_0, f_1, \dots, f_m$  are all convex functions.

### Optimality Cond. for differentiable $f_0$

If  $f_0$  is diff. then:

$$f_0(y) - f_0(x) \geq \nabla^T f(x)[y - x]$$

If  $X$  denotes the feasible set i.e.,

$$\{x \mid f_i(x) \leq 0, \quad i \in [m], \quad Ax = b\}$$

Then  $x^*$  is optimal iff  $\forall y \in X$ :

$$\nabla^T f(x^*)[y - x^*] \geq 0 \quad \forall y \in X$$

*Proof.* ( $\Leftarrow$ ) :

$$\nabla^T f(x^*)[y - x^*] \geq 0 \quad \forall y \in X$$

Then  $f_0(y) \geq f_0(x^*) + \nabla^T f(x^*)[y - x^*] \geq f_0(x^*)$ .  $\therefore x^*$  is opt.

( $\Rightarrow$ ) : If  $x^*$  is optimal. Suppose  $\nabla^T f(x^*)(y - x^*) < 0$  for some  $y \in X$ . Then  $X$  is convex, so  $z(t) = ty + (1 - t)x^*$  is in  $X$  for  $t \in [0, 1]$ .

$$\left. \frac{d}{dt} f_0(z(t)) \right|_{t=0} = \nabla^T f(x^*)(y - x^*) < 0$$

$\implies$  decreasing in nbd. of 0.  $\square$

## Optimization with Equality Constraints

Consider the optimization problem:

$$\begin{aligned} \min \quad & f_0(x) \\ \text{subject to} \quad & Ax = b \end{aligned}$$

where  $f_0$  is differentiable.

The first-order optimality condition for a feasible point  $x$  is:

$$\nabla f_0(x)^T (y - x) \geq 0 \quad \forall y \in \{y \mid Ay = b\}$$

Let  $v = y - x$ . Since both  $x$  and  $y$  are feasible, we have  $A(y - x) = A(x) - A(y) = 0$ , which implies  $v \in \mathcal{N}(A)$  (the null space of  $A$ ). The condition is equivalently stated as:

$$\nabla f_0(x)^T v \geq 0 \quad \forall v \in \mathcal{N}(A)$$

**Lemma 6.1.** Let  $f : V \rightarrow \mathbb{R}$  be a linear functional on a subspace  $V$ . If  $f(v) \geq 0$  for all  $v \in V$ , then  $f(v) = 0$  for all  $v \in V$ .

*Proof.* Since  $V$  is a subspace, for any  $v \in V$ , its additive inverse  $-v$  is also in  $V$ . By the linearity of  $f$ :

$$f(-v) = -f(v)$$

By the hypothesis, we must have  $f(v) \geq 0$  and  $f(-v) \geq 0$ . This implies:

$$f(v) \geq 0 \quad \text{and} \quad -f(v) \geq 0 \implies f(v) = 0$$

Thus,  $f \equiv 0$  on  $V$ . □

Applying this Lemma to our optimality condition, we have:

$$\nabla f_0(x)^T v = 0 \quad \forall v \in \mathcal{N}(A)$$

This implies that the gradient is orthogonal to the null space of  $A$ :

$$\nabla f_0(x) \perp \mathcal{N}(A)$$

Recall the fundamental theorem of linear algebra:  $\mathcal{N}(A)^\perp = \mathcal{C}(A^T)$ . Therefore,  $\nabla f_0(x)$  must lie in the range of  $A^T$ . This means there exists a vector of Lagrange multipliers  $\nu \in \mathbb{R}^m$  such that:

$$\nabla f_0(x) + A^T \nu = 0$$

## 7 Duality in Convex Optimization

$$\begin{aligned} & \min f_0(x) \\ & \text{subject to } f_i(x) \leq 0 \quad i \in [m] \\ & \quad \quad h_i(x) = 0 \quad i \in [p] \end{aligned}$$

Define Lagrangian  $L : \mathbb{R}^n \times \mathbb{R}^m \times \mathbb{R}^p \rightarrow \mathbb{R}$

$$L(x, \lambda, \nu) = f_0(x) + \sum_{i=1}^m \lambda_i f_i(x) + \sum_{i=1}^p \nu_i h_i(x)$$

**Lagrange Dual fn:**

$$g(\lambda, \nu) = \inf_{x \in \mathcal{D}} L(x, \lambda, \nu)$$

**Proposition 7.1.**  $\forall \lambda \geq 0, \nu$ ,

$$g(\lambda, \nu) \leq p^*$$

where  $p^*$  is optimum value for (i).

*Proof.* Let feasible set for (i) be denoted by  $X$ . If  $x \in X$  then  $f_i(x) \leq 0$  and  $h_i(x) = 0$ .

$$L(x, \lambda, \nu) \leq f_0(x) \quad \forall x \in X$$

$$\inf_{x \in \mathcal{D}} L(x, \lambda, \nu) \leq \inf_{x \in X} f_0(x) = p^*$$

$$\implies g(\lambda, \nu) \leq p^* \quad \forall \lambda \geq 0, \nu. \quad \square$$

**Dual problem: of (i)**

$$\max g(\lambda, \nu)$$

$$\text{sub. to } \lambda \geq 0$$

**Weak Duality:** If  $\max_{\lambda \geq 0} g(\lambda, \nu) = d^*$  Then  $d^* \leq p^*$ .

**Remark:** Note that, till now, for Duality, we did not use that our optimization problem is for convex optimization, so, Weak Duality still holds for non-convex optimizatn problems.

## Strong Duality & Slater's Constraint

If  $d^* = p^*$  holds, then we call that **strong duality** (like in LPP).

### Some more notions

**Definition 7.1. Affine Set:**  $C$  is an affine set if  $\forall x, y \in C$  and  $\forall \theta \in \mathbb{R}$ ,

$$\theta x + (1 - \theta)y \in C$$

**Definition 7.2. Affine Hull:** The smallest affine set containing set  $C$ .

**Definition 7.3. Relint:** Interior wrt affine hull (relative interior).

$$\text{relint}(C) = \{x \in C \mid \exists \epsilon > 0 \text{ s.t. } N_\epsilon(x) \cap \text{aff}(C) \subseteq C\}$$

### Slater's Condition

$\exists x \in \text{relint}(\mathcal{D})$  such that

$$f_i(x) < 0, \quad i = 1, \dots, m, \quad Ax = b$$

Such a point is called **strictly feasible**.

**Theorem 7.1. Slater's Thm:** If Slater's cond. holds, then **strong duality** holds.

*Proof.* For simplicity take  $A$  to be full row rank and  $\text{relint}(\mathcal{D}) = \text{int}(\mathcal{D})$ , let the strictly feasible point be denoted by  $\tilde{x}$

Take  $S, T \subseteq \mathbb{R}^m \times \mathbb{R}^p \times \mathbb{R}$  defined as follows :

$$S = \{(u, v, t) \mid f_i(x) \leq u_i, h_j(x) = v_j, f_0(x) \leq t \text{ for some } x \in \mathcal{D}, \forall i \in [m], j \in [p]\}$$

$$T = \{(0, 0, t) \mid t < p^*\}$$

Now note that  $S, T$  are clearly convex. Claim :  $S \cap T = \emptyset$  Proof: Let  $(u, v, t) \in S \cap T \implies u = \underline{0}, v = \underline{0}, t < p^*$  again as it is in  $S$  so, we have for some  $x \in \mathcal{D}, \forall i, j, f_i(x) \leq 0, h_j(x) = 0 \implies x$  is feasible solution and so  $f_0(x) \geq p^*$

And so,  $p^* \leq f_0(x) \leq t \leq p^*$  and hence contradiction.

Thus we get  $S \cap T = \emptyset$  and so by Separating Hyper-plane Theorem we can say that  $\exists (\lambda_0, \nu_0, t_0) \neq \underline{0} \in \mathbb{R}^m \times \mathbb{R}^p \times \mathbb{R}$  and  $\alpha \in \mathbb{R}$  such that  $(\lambda_0, \nu_0, t_0)^T (u, v, t) \geq \alpha \forall (u, v, t) \in S \dots (*)$  and  $(\lambda_0, \nu_0, t_0)^T (u, v, t) < \alpha \forall (u, v, t) \in T \dots (**)$

Now we claim that  $\lambda_0 \geq 0$  and  $t_0 \geq 0$ . Otherwise if  $\lambda_0 < 0$  or  $t_0 < 0$ , Note that making  $u_i$  s arbitrarily large we can make the lhs of  $(*)$  unbounded below, givingus a contradiction.

Now from  $(**)$  we get that  $\forall t < p^*, t_0 t < \alpha \implies t_0 p^* \leq \alpha$

For  $x \in \mathcal{D}$ , from  $*$  we get that  $\sum_{i=1}^m \lambda_{0i} f_i(x) + \nu_0^T [Ax - b] + t_0 f_0(x) \geq \alpha \geq t_0 p^*$ . If  $t_0 = 0$  Then Taking  $x = \tilde{x}$ , we get that  $\sum_{i=1}^m \lambda_{0i} f_i(\tilde{x}) \geq 0 \implies \lambda_0 = \underline{0}$  as  $f_i(\tilde{x}) < 0 \forall i$  So, we have for all  $x \in \mathcal{D}$ ,

$$\nu_0^T (Ax - b) \geq 0 \dots (***)$$

$\nu_0 \neq \underline{0}$ . Now note that  $\nu_0^T (A\tilde{x} - b) = 0$ . Now,  $\tilde{x}$  is an interior point of  $\mathcal{D}$  and hence for some  $\epsilon > 0$ , we have  $N_\epsilon(\tilde{x}) \subseteq \mathcal{D}$ . Take  $v \in N_\epsilon(\tilde{x}) \setminus \{0\}$ . Now note that

$$\|\tilde{x} - (\tilde{x} - tv)\| = |t| \cdot \|v\|$$



So, for some  $\delta > 0$ , we have  $\forall t \in (-\delta, \delta)$ ,  $x + tv \in \mathcal{D}$ . Now note that  $\nu'_0(A(\tilde{x} + tv) - b) = t\nu'_0Av$ . From (\*\*\*) , we get that  $\nu_0Av = 0$  But note that  $A$  is full row rank and  $nu_0 \neq 0$  so  $\rho(\nu'_0A) = 1$ . Thus contradiction as  $v \neq 0$  Thus  $t_0 \neq 0$

Hence, By diving by  $t_0$  both sides, we get that

$$\begin{aligned} \sum_{i=1}^m \frac{\lambda_{0i}}{t_0} f_i(x) + \left( \frac{\nu_0}{t_0} \right)^T [Ax - b] + f_0(x) &\geq p^* \quad \forall x \in \mathcal{D} \\ \implies \inf_{x \in \mathcal{D}} L(x, \frac{\lambda_0}{t_0}, \frac{\nu_0}{t_0}) &\geq p^* \\ \implies g(\frac{\lambda_0}{t_0}, \frac{\nu_0}{t_0}) &\geq t^* \\ \implies \max_{\lambda \geq 0, \nu} g(\frac{\lambda_0}{t_0}, \frac{\nu_0}{t_0}) &\geq t^* \implies d^* \geq p^* \end{aligned}$$

Thus  $\boxed{d^* = p^*}$

□

**Lemma 7.1** (Separating Hyperplane Theorem). ..

## 8 Non-Convex Optimization, KKT Conditions

In the following section, we **don't** assume that the optimization problem : Minimize  $f_0(x)$  subject to  $f_i(x) \leq 0 \quad i = 1, \dots, m$  , and  $Ax = b$ , is necessarily convex.

### Suboptimality

Note that  $p^* \geq g(\lambda, \nu) \quad \forall \lambda \geq 0$  and so, if  $x$  is primal feasible and  $(\lambda, \nu)$  is dual feasible then we have  $f(x) - g(\lambda, \nu) \geq f(x) - p^*$ . This is called the **Duality Gap** associated with primal feasible  $x$  and dual feasible  $(\lambda, \nu)$ .

### Complementary Slackness

Let  $x^*$  be a primal optimal and  $(\lambda^*, \nu^*)$  be the dual optimal Then we must have  $f(x^*) \geq g(\lambda^*, \nu^*)$ . If we further assume that strong duality holds then we have  $f(x^*) = g(\lambda^*, \nu^*) = \inf_{x \in \mathcal{D}} L(x, \lambda^*, \nu^*) \leq L(x^*, \lambda^*, \nu^*) = f_0(x^*) + \sum_{i=1}^p \lambda_i^* f_i(x^*) + \sum_{j=1}^m \nu_j^* h_j(x^*) = f_0(x^*) + \sum_{i=1}^p \lambda_i^* f_i(x^*)$  From here we get that

$\sum_{i=1}^p \lambda_i^* f_i(x^*) = 0$ . This is known as complementary slackness. Moreover from the above inequality we can also say that if strong duality holds then  $g(\lambda^*, \nu^*) = L(x^*, \lambda^*, \nu^*)$

### KKT optimal condition

Let our function  $f_0, f_i, h$  all are differentiable. Then note that If strong duality holds then,  $L(x^*, \lambda^*, \nu^*)$  is the maximizer of the dual problem so, the gradient of  $L$  at that point must be zero. This is known as the KKT conditions for optimization problems.

## 9 Support Vector Classifier and Support Vector Machines

### Linearly Separable Case: Maximal Margin SVC

First consider the linearly separable case. We have two sets of observations that have come from two classes. As an example, may have been generated from two Normal Distribution. In linearly separable case, we consider that there is overlap between the two groups. Now we want to separate them by a hyperplane. Now note that, if a hyperplane indexed by the vectors  $b, b_0$  are able to separate the data into two classes completely, then by tweaking them a little bit, we can also do the job. So, there are infinitely many such separating hyperplane. So, which one is the best ?

We solve this problem by introducing a concept called Margin, the minimum distance between the separating hyper-plane and the observations, and we want to maximize this quantity. Intuitively, it can be thought like, this is the maximal "slab" we can insert between the two classes of observations. This Classifier is called the Maximal Margin SVC. ( Why the word "support vector", that we will see later.)

### Formalizing the problem

So, the above discussion can be seen as an optimization problem. Say, the training observations are  $\{\underline{x}_i, y_i\}_{i=1}^n$  where  $y_i \in \{1, -1\}$ . So , we want a hyper-plane indexed by the scalar and the vector  $b_0, b$  respectively. And we want that  $x'_i b + b_0 > 0$  if  $y_i = 1$  and  $x'_i b + b_0 < 0$  if  $y_i < 0$ . Equivalently, we want that  $y_i(x'_i b + b_0) > 0 \forall i$ . And if we take  $b$  to be a unit vector then we also want that  $\min_i y_i(x'_i b + b_0)$  is maximized so, the problem is equivalent to the optimization problem:

$$\begin{aligned} & \max_{||b||=1, b_0} M \\ & \text{subject to } y_i(x'_i b + b_0) \geq M \forall i \end{aligned}$$

equivalently,

$$\begin{aligned} & \min_{b, b_0} ||b|| \\ & \text{subject to } y_i(x'_i b + b_0) \geq 1 \forall i \end{aligned}$$

### Linearly inseparable case : Soft- margin SVC

Now we shall deal with when the two classes have overlaps and can't be separated linearly in the feature space ( we may transform them into some other space to make them linearly separable, more on that later) Then we allow some observations, inside the margin, even in the wrong side of the hyperplane but we will not make many of this misclassifications naturally.

Mathematically , we will introduce some slack variables,  $\xi_i$ , which will denote the relative distance of it from the right side of the margin. Now if  $\xi_i > 1$  then it is a misclassification. We want to bound number of misclassification by a constant hence, a sensible constraint is to put a bound on the quantity  $\sum_{i=1}^n \xi_i$  So in this case our problem is

$$\begin{aligned} & \min_{b, b_0} ||b|| \\ & \text{subject to } \xi_i \geq 0, \xi_i \leq \text{constant}, y_i(x'_i b + b_0) \geq (1 - \xi_i) \end{aligned}$$

To make this problem to a convex optimization problem we take the following set-up :

$$\begin{aligned} & \min_{b, b_0} \frac{1}{2} ||b||^2 + C \sum_{i=1}^n \xi_i \\ & \text{subject to } \xi_i \geq 0, \\ & y_i(x'_i b + b_0) \geq (1 - \xi_i) \end{aligned}$$

### Fitting the SVC

So, our optimization problem is :

$$\begin{aligned} & \min_{b, b_0, \xi} \frac{1}{2} ||b||^2 + C \sum_{i=1}^n \xi_i \\ & \text{subject to } \xi_i \geq 0, \\ & y_i(x'_i b + b_0) \geq (1 - \xi_i) \end{aligned}$$

Where  $C$  is the cost parameter and for linearly separable case we may consider  $C \rightarrow \infty$   
The primal Lagrangian is :

$$L_p(b, b_0, \xi, \lambda, \nu) = \frac{1}{2} \|b\|^2 + C \sum_{i=1}^n \xi_i - \sum_{i=1}^n \lambda_i \xi_i - \sum_{i=1}^n \nu_i (y_i(x'_i b + b_0) - (1 - \xi_i))$$

Note that as our objective function is strongly convex ( defined after) so our objective function will have a unique minimizer.

Now the dual function is :  $\min_{b, b_0, \xi} L_p = L_D(\lambda, \nu)$  So, taking the gradient to be zero, we get that

$$\begin{aligned} b &= \sum_{i=1}^n \nu_i y_i x_i \\ \sum_{i=1}^n \nu_i y_i &= 0 \\ C - \nu_i - \lambda_i &= 0 \end{aligned}$$

Now we take the dual problem ( this is actually the wolfe dual but similar to the lagrangian theory)  
maximize  $L_D(\lambda, \nu) = \sum \nu_i - \frac{1}{2} (\sum \nu_i y_i x_i)' (\sum \nu_i y_i x_i)$   
subject to  $C \geq \nu \geq 0, \sum \nu_i y_i = 0$

From KKT conditions, ( complementary slackness) we also get that

$$\begin{aligned} \lambda_i \xi_i &= 0 \\ \nu_i (y_i(x'_i b + b_0) - (1 - \xi_i)) &= 0 \\ y_i(x'_i b + b_0) &\geq (1 - \xi_i) \end{aligned}$$

Now, we have solutions that  $\hat{b} = \sum \hat{\nu}_i y_i x_i$

Note that this only depend on those  $x_i$  for which the inequality  $y_i(x'_i b + b_0) \geq (1 - \xi_i)$  is attaining equality and our optimal hyper plane is then  $\hat{b}'x + \hat{b}_0$ .

The vectors for which  $y_i(\hat{b}'x_i + \hat{b}_0) = 1 - \hat{\xi}_i$ , only contribute to  $\hat{b}$ , they are called support vectors. Among this some will be on the margin, for which  $\hat{\xi}_i = 0$  or equivalently  $\nu_i \in (0, C)$ , For other points we will have,  $\hat{\xi}_i > 0 \implies \hat{\nu}_i = C$ . We can solve for  $b_0$  for any margin point characterized by  $\nu_i \in (0, C)$  The vectors which are not support vectors, deeper points, for them automatically ,  $\hat{\xi}_i = 0$ .

## Supprt Vector Machines: Why Care about Kernels?

Note that we have the  $b = \sum \nu_i y_i x_i$  and  $b_0$  can be solved by solving  $y_i(x'_i b + b_0) = 1$  for any  $i$  with  $\nu_i \in (0, C)$ .

Then , if we use under some transformation  $h$  the same thing we will get the Dual to be  $L_D = \sum \nu_i - (\sum \nu_i y_i h(x_i))' (\sum \nu_i y_i h(x_i))$ . Now, maximize this subject to  $\nu_i \in [0, C]$  and  $\sum \nu_i y_i = 0$ . We will get  $\hat{\nu}_i$ .

our  $\hat{b}$  will be of the form  $\sum \hat{\nu}_i y_i h(x_i)$ . Again only the  $i$  s for which  $\nu_i \neq 0$  will contribute. For some lying on the margin,  $\nu_i \in (0, C)$ . Solving for any one of them we can get  $\hat{b}_0$ , for others slack variable will be positive and so ,  $\hat{\nu}_i = C$ .

But what is important is that note that  $L_D$  contains a dot product like  $\sum \sum \nu_i \nu_j y_i y_j \langle h(x_i), h(x_j) \rangle$  and the resulting hyperplane in the transformed feature space is also  $\sum \hat{\nu}_i y_i \langle h(x_i), h(x) \rangle + \hat{b}_0$  . So we don't need actually the transformation function, all we need is a valid candidate for inner product function  $K(x, x')$

### Kernels used

Kernels used majorly are :

1.  $d$ -degree polynomial kernel  $= (1 + \langle x, x' \rangle)^d$
2. Radial Kernel :  $\exp(-\gamma \|x - x'\|_2^2)$

### Why do we bother ?

In the tranfomed space the inputs can be linearly separable and more in the original feature space we get a non-linear decision boundary leading to more robustness.

## 10 Convex Clustering

### Clustering Centroids Optimization

Given  $n$  data pts  $X = \{x_1, \dots, x_n\} \subseteq \mathbb{R}^p$ , we seek cluster centroids  $u_i \in \mathbb{R}^p$  for each  $x_i$  to:

$$\min_{u \in \mathbb{R}^{np}} E_\gamma(u) = \frac{1}{2} \sum_{i=1}^n \|x_i - u_i\|_2^2 + \gamma \sum_{(i,j) \in \mathcal{E}} w_{ij} \|u_i - u_j\|_2$$

Where  $\gamma$  is the **tuning parameter** the second term is the lasso-type penalty for complexity and  $w_{ij} \geq 0$  are weights denoting similarity  $i$  th and  $j$ th observation. And,  $\mathcal{E} = \{(i, j) \in [n]^2 | w_{ij} > 0\}$

### Strong Convexity

**Definition 10.1. Strong Convexity:** A twice diff. convex function  $f : \mathbb{R}^n \rightarrow \mathbb{R}$  is called **strongly convex** on  $S \in \mathbb{R}^n$  if it grows faster than the quadratic, i.e.,

$$\nabla^2 f(x) \succeq mI \quad \text{for some } m > 0.$$

(difference of two matrices is an nnd)

Recall the Taylor's Theorem. If  $f$  is doubly differentiable then  $f(y) = f(x) + \nabla^T f(x)(y - x) + \frac{1}{2}(y - x)^T \nabla^2 f(x)(y - x)$ . So, the next proposition follows from the non-negative definiteness of the difference matrix.

**Proposition 10.1.** *If  $f$  is strongly convex, then:*

$$f(y) \geq f(x) + \nabla f(x)^T (y - x) + \frac{m}{2} \|y - x\|_2^2$$

### Strong Convexity Properties

**Lemma 10.1.** If  $f$  is strongly convex, then the optimal point  $x^*$  is unique.

*Proof.*  $\forall y \in S, f(y) \geq f(x) + \nabla f(x)^T (y - x) + \frac{m}{2} \|y - x\|_2^2$ .

RHS is a quadratic function for  $y$  for fixed  $x$  and the minimum occurs for  $\tilde{y} = x - \frac{1}{m} \nabla f(x)$ . Substituting gives  $p^* \geq f(x) - \frac{1}{2m} \|\nabla f(x)\|_2^2$ . If  $x^*$  is optimal,  $\|x^* - x\|_2^2 \leq \frac{2}{m} \|\nabla f(x)\|_2^2$ .

If  $\exists x_1$  another optimal point, then  $\nabla f(x_1) = 0$ , so  $\|x^* - x_1\|_2^2 \leq 0 \implies x_1 = x^*$ .  $\square$

**Lemma 10.2.** If  $f$  is strongly convex and  $g$  is convex then  $f + g$  is strongly convex.

*Proof.* Easy, skipped  $\square$

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**Remark:** Note that for  $E_\gamma(u)$ , the first term is strongly convex and the second term is convex making the function strongly convex. And hence for every fixed  $\gamma$ , there is a unique minimizer  $u(\gamma)$  and hence unique centroids  $u_i(\gamma)$  for all  $i$ 's.

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## Clustering Centroids: The Role of $\gamma$

Why bother about  $\gamma$ ?

- When  $\gamma = 0$ :  $u_i = x_i$ . Each point is getting its own cluster.
- As  $\gamma$  increases from 0: Clusters fuse together.  $\gamma$  trades off the emphasis between data fit and pairs of centroids.
- When  $\gamma$  is very large:  $u_i$  is the center of mass (COM) of  $X$ .

Let  $\mathcal{P} = \{P_1, \dots, P_k\}$  denote a partition of  $X$  inducing index partition  $\mathcal{I}$ .

### Weights

$w_{ij} \rightarrow$  intuitively measuring similarities between  $x_i$  and  $x_j$ . If  $i, j$  are "close", then their centroids should not be far, so a greater penalty should be imposed on that.

Consider the graph  $G = ([n], \mathcal{E})$ . If it has  $k$  connected components inducing partition  $\mathcal{I}_1, \dots, \mathcal{I}_k$ , then:

$$E_\gamma(u) = \sum_{j=1}^k E_\gamma^j(u)$$

$$E_\gamma^j(u) = \sum_{i \in \mathcal{I}_j} \|x_i - u_i\|_2^2 + \gamma \sum_{(i,j) \in \mathcal{E}_k} w_{ij} \|u_i - u_j\|_2$$

Intuitively, as  $\gamma$  increases,  $E_\gamma^k(u)$  should converge to the COM of the  $k$ -th partition, i.e.,  $\frac{1}{|\mathcal{I}_k|} \sum_{i \in \mathcal{I}_k} x_i$ . (Analytic proof skipped)

Consider edges between 2 connected components of the graph. Let  $k = 2$ , with  $\mathcal{I}_1, \mathcal{I}_2$  being index partitions. Now,  $\mathcal{E}_{12} = \{(i, j) : i \in \mathcal{I}_1, j \in \mathcal{I}_2\}$ . The modified criterion is:

$$E_\gamma(u) = E_\gamma^1(u) + E_\gamma^2(u) + \gamma \sum_{(i,j) \in \mathcal{E}_{12}} w_{ij} \|u_i - u_j\|_2$$

If  $w_{ij}$  for  $(i, j) \in \mathcal{E}_{12}$  is very small compared to within-cluster weights, then we can expect as  $\gamma$  increases, first local averaging of centering will occur and then eventually  $\mathcal{I}_1$  and  $\mathcal{I}_2$  centroids will fuse.

### Weights in Practice

- **Inverse Euclidean Norm:**  $w_{ij} = \frac{1}{\|x_i - x_j\|_2}$
- **Gaussian Kernel:**  $w_{ij} = \exp\left(-\frac{\|x_i - x_j\|_2^2}{\sigma_{ij}}\right)$ , where  $\sigma_{ij} = \sigma_i \cdot \sigma_j$ .

$\sigma_i$  is a local measure of dispersion around  $x_i$ , e.g., the median Euclidean distance between  $x_i$  and its nearest neighbors.

### Perfect Recovery

Perfect Recovery means to select a suitable tuning parameter  $\gamma$  so that the algorithm correctly identifies the underlying cluster structure.